

EEE3097/8/9S Micromouse 2026: Design Report Guidelines

Graduate Attribute 3 (Design) Assessment Portfolio

1. Overview

As part of the course assessment, you must submit two formal engineering design reports documenting a structured design process conducted in relation to the Micromouse project.

The main purpose of these submissions is to demonstrate proficiency in the **ECSA Design Graduate Attribute (GA3)**, which is a mandatory requirement to pass the course.

- **Design Report 1 (25%):** Submitted alongside the Milestone 1 assessment.
 - **Design Report 2 (30%):** Submitted alongside the Milestone 2 assessment.
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2. Format & Submission Rules

To ensure standardization for external accreditation audits, you must adhere to the following formatting constraints:

* **Template:** You must use the Microsoft Word form template provided at EEE3097_8_9S_designreport.docx. *

Form Protection: Do not disable the “Protect Form” functionality in MS Word. The template enforces strict layout limits and character constraints for each section. * **Output:** Convert the finalized document to **PDF** for submission to Gradescope (named `Design_Report_[1/2]_[StudentNumber].pdf`).

3. Open-Ended Design Task Selection

You can choose **any design task** from your Micromouse development for which you can confidently present evidence of design thinking.

Without being prescriptive, some illustrative examples of design tasks you could document include: 1. **Dynamic Simulation Modeling:** Designing a more realistic and operationally useful Simulink model or Python physics representation of the Micromouse DC motor drive and friction system. 2. **Closed-Loop Speed Controller:** Designing, tuning, and discretizing a PID feedback controller to match wheel speeds. 3. **Robust Heading & Yaw Alignment:** Fusing gyroscope rates and encoders to steer the mouse and correct drift. 4. **Active Wall Centering:** Fusing front and side ToF sensor readings to steer down the corridor center. 5. **Algorithmic Path Exploration:** Designing Stateflow state machines, algorithmic flowcharts, or Python FSM classes to explore and map the maze cells. 6. **Pathfinding Optimizations:** Formulating Floodfill or BFS algorithms to calculate the shortest path. 7. **Educational Interfaces (Blockly):** Designing a child-friendly visual block programming library and Python code generator. 8. **Watchdog Safety Wrappers:** Designing software safety blocks or safety overrides to prevent collisions.

4. Required Report Sections & Strict Constraints

To encourage concise, high-density engineering communication and prevent AI-generated filler, strict character limits are enforced (including spaces). Your report must populate the template sections, detailing: * **Brief Description of Design Task (Max 350 chars):** Outline the subsystem task and the problem it solves. * **Visual Aids:** Insert a single graphic (e.g. block diagram, Stateflow transition chart, or telemetry plot) referenced in the text. *

Design Criteria & Constraints (Max 700 chars): Define quantitative target metrics and physical/computational limits. * **Main Design Assumptions (Max 700 chars):** Outline and mathematically justify your engineering assumptions. * **Design Process Description (Max 700 chars):** Document your structured evaluation of at least two alternative solutions. * **Design Implementation (Max 700 chars):** Model your chosen design using first-principles dynamic equations or script flow charts. * **Design Evaluation & Testing (Max 700 chars):** Verify performance under physical perturbations (slip, motor asymmetry) and discuss model deviations.